

School of Computer Sciences
Department of Computer Science

Name of the Academic Program: Master of Computer Applications

Course Code: **PCACC20201**

Title of the Course: **Operating System**

L-T-P: 4-0-2 Credits: 5

Prerequisite Course / Knowledge (If any):

- The students should hold basic knowledge of computer System.
- The students should have practical knowledge of working with computers.

Course Outcomes (COs)

After completion of this course successfully, the students will be able to

CO-1: Review the basic concepts of operating system. (Level 2: Understand)

CO-2: Illustrate the examples on processor scheduling and deadlock prevention. (Level 3: Apply)

CO-3: Justify the demand paging concepts for the comparison of page replacement algorithms.(Level 5: Evaluate)

CO-4: Describe the disk scheduling algorithms and its comparison in terms of performance. (Level 5: Evaluate)

CO-5: Review the file system and its protection mechanism. (Level 2: Understand)

Mapping of Course Outcomes (COs) with Program Outcomes (POs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	1	--	2	--	2	1	--	--	--
CO2	3	3	3	2	--	2	2	--	1	--
CO3	3	3	3	1	--	--	1	--	1	--
CO4	3	2	2	1	--	--	1	--	2	--
CO5	2	2	1	--	--	--	1	--	1	--

PCACC20201:Operating System

Credits: 5

IA: 50 Marks

Contact Hrs (L:T:P): 6 (4:0:2)

End Exam: 75 Marks

Unit 1

(15 hrs)

Introduction to Operating System: Operating systems concepts, types of operating systems, Evolution of operating systems, different views of the operating system, operating system services, System calls, Types of system calls, operating system structure, layered approach, Micro kernels and Virtual machines.

Unit 2

(15 hrs)

Process Management: Process concept, operation on processes, Inter-process communication, mutual exclusion, process synchronization, Inter process synchronization, critical section problem, semaphores, process scheduling concepts, scheduling criteria, scheduling algorithms, deadlocks, system model, deadlock characterization, deadlock prevention, deadlock avoidance.

Unit 3

(15 hrs)

Memory Management: Introduction, memory management, swapping, contiguous memory allocation, paging, segmentation, virtual memory, demand paging, page replacement algorithms: FIFO, Optimal, LRU, Counting based page replacement.

Unit 4

(15 hrs)

Disk Scheduling: Introduction, physical characteristics, disk scheduling algorithms, disk Management, RAID Structure.

File system: Files, access method, directory structure, protection and file system implementation, allocation methods.

Protection: Goals, mechanism and policies, domain of protection, access matrix and its implementation, dynamic protection structure, revocation, security.

Reference Books:

1. J. Sliberschatz (2006), *Operating systems Concepts*, McGraw Hill.
2. Madnick, S.E. Donovan J.J. (1974), *Operating system*, McGraw Hill.
3. Brinch Hansen P (1973), *Operating system Principles*, PHI.
4. Milan Milenkovic (2001), *Operating systems*, McGraw Hill.
William Stallings (2018), *Operating system Internals and Design Principles*, Pearson.